Ansible Automation - Use Cases

A reference of the use cases by which Ansible Automation Platform is commonly implemented, by domain. Each use case is described in the same format (Description, Collection, Key content, Notes) and mapped to Red Hat Certified and Validated content.

3 solution use cases · 14 product use cases · generated 2026-10-21 · community reference, not an official Red Hat publication

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Other clouds (IBM, Oracle, OpenStack...)

IBM Cloud, Oracle Cloud, OpenStack and other cloud platforms.

Solution use cases

Use Case: Provision Oracle Cloud (OCI) resources

Description	Create an Oracle Cloud Infrastructure environment with a playbook: compartments, a virtual cloud network, subnets, security lists, compute instances and a load balancer. The same job template builds each environment the same way, and the inventory plugin finds the new instances afterwards.
Collection	`ansible.oci` (Certified)
Key content	• `ansible.oci.oci_identity_compartment` — Manage OCI compartments • `ansible.oci.oci_network_vcn` — Manage a Virtual Cloud Network resource in Oracle Cloud Infrastructure • `ansible.oci.oci_network_subnet` — Manage a Subnet resource in Oracle Cloud Infrastructure • `ansible.oci.oci_network_security_list` — Manage a Security List resource in Oracle Cloud Infrastructure • `ansible.oci.oci_instance` — Manage a Compute instance resource in Oracle Cloud Infrastructure • `ansible.oci.oci_loadbalancer` — Manage an OCI load balancer • `ansible.oci.oci_compute` — Oracle Cloud Infrastructure Compute inventory source
Notes	• Protect the data with Back up Oracle Cloud block and boot volumes. • Keep the OCI API private key in an Automation Platform credential, not in the project. • For Terraform-managed OCI estates, see Run Terraform from Automation Platform.

Use Case: Back up Oracle Cloud block and boot volumes

Description	Make sure every OCI block volume, boot volume and volume group has a backup schedule, and take extra backups before maintenance. Policies are defined in code and assigned to volumes automatically, so new volumes are not left unprotected.
Collection	`ansible.oci` (Certified)
Key content	• `ansible.oci.oci_volume_backup_policy` — Manage a volume backup policy in Oracle Cloud Infrastructure • `ansible.oci.oci_volume_backup_policy_assignment` — Manage an OCI volume backup policy assignment • `ansible.oci.oci_volume_backup` — Manage a block volume backup resource in Oracle Cloud Infrastructure • `ansible.oci.oci_boot_volume_backup` — Manage a boot volume backup resource in Oracle Cloud Infrastructure • `ansible.oci.oci_volume_group_backup` — Manage a volume group backup resource in Oracle Cloud Infrastructure

Notes	• Keep restore tests in a separate compartment to avoid conflicts with production names and quotas. • Related: Provision Oracle Cloud (OCI) resources.

Use Case: Migrate OpenStack tenants and workloads

Description	Move a tenant from one OpenStack cloud to another, for example during a platform upgrade or consolidation. The collection exports the tenant's resources to files, recreates them in the destination, and copies instance data through conversion hosts.
Collection	`os_migrate.os_migrate` (Certified)
Key content	• `os_migrate.os_migrate.deploy_conversion_hosts` — Deploy conversion hosts • `os_migrate.os_migrate.export_networks` — os-migrate resource • `os_migrate.os_migrate.import_networks` — os-migrate resource • `os_migrate.os_migrate.export_workloads` — os-migrate resource • `os_migrate.os_migrate.import_workloads` — os-migrate resource • `os_migrate.os_migrate.validate_data_dir` — os-migrate resource • `os_migrate.os_migrate.delete_conversion_hosts` — Delete conversion hosts
Notes	• Migrate in waves, starting with a low-risk tenant, and keep the source running until the destination is verified. • Run the export and import steps from a workflow so each step is logged and repeatable. • Plan for downtime of each instance during its final volume copy.

Product use cases

Grouped from the modules of each certified or validated collection. Key content lists the components each use case supports, with the modules that manage them (create, update, delete) and query them (_info).

Oracle Cloud Infrastructure `ansible.oci`

Certified Red Hat · version 1.1.0 · updated 2026-10-06 · 102 modules → 8 use cases

Use Case: Manage compute instances and images

Description	Launch and manage Compute instances, custom images, console connections and VNIC attachments, look up shapes, and build inventory from OCI Compute.
Collection	`ansible.oci` (Certified, Red Hat)
Key content	**Compute instances** `oci_instance`, `oci_instance_info` **Custom images** `oci_image`, `oci_image_info` **Compute shapes** `oci_shape_info` **Console connections** `oci_instance_console_connection`, `oci_instance_console_connection_info` **VNIC attachments** `oci_vnic_attachment`, `oci_vnic_attachment_info` **Compute inventory** inventory `oci_compute`

Use Case: Manage block and boot volumes and backups

Description	Create and attach block and boot volumes and volume groups, take backups, and assign backup policies.
Collection	`ansible.oci` (Certified, Red Hat)
Key content	**Block volumes** `oci_blockstorage_volume`, `oci_blockstorage_volume_info` **Volume attachments** `oci_volume_attachment`, `oci_volume_attachment_info` **Boot volumes** `oci_boot_volume`, `oci_boot_volume_info` **Boot volume backups** `oci_boot_volume_backup`, `oci_boot_volume_backup_info` **Volume backups** `oci_volume_backup`, `oci_volume_backup_info` **Volume groups** `oci_volume_group`, `oci_volume_group_info` **Volume group backups** `oci_volume_group_backup`, `oci_volume_group_backup_info` **Backup policies** `oci_volume_backup_policy`, `oci_volume_backup_policy_info` **Backup policy assignments** `oci_volume_backup_policy_assignment`, `oci_volume_backup_policy_assignment_info`

Use Case: Build virtual cloud networks

Description	Create VCNs, subnets, route tables, gateways, DHCP options and IP addresses, and connect networks through DRGs and peering.
Collection	`ansible.oci` (Certified, Red Hat)
Key content	**Virtual cloud networks** `oci_network_vcn`, `oci_network_vcn_info` **Subnets** `oci_network_subnet`, `oci_network_subnet_info` **Route tables** `oci_network_route_table`, `oci_network_route_table_info` **DHCP options** `oci_network_dhcp_options`, `oci_network_dhcp_options_info` **Internet gateways** `oci_network_internet_gateway`, `oci_network_internet_gateway_info` **NAT gateways** `oci_network_nat_gateway`, `oci_network_nat_gateway_info` **Service gateways** `oci_network_service_gateway`, `oci_network_service_gateway_info` **Local peering gateways** `oci_network_local_peering_gateway`, `oci_network_local_peering_gateway_info` **Dynamic routing gateways** `oci_network_drg`, `oci_network_drg_info` **DRG attachments** `oci_network_drg_attachment`, `oci_network_drg_attachment_info` **Private IPs** `oci_network_private_ip` **Public IPs** `oci_network_public_ip`

Use Case: Secure network traffic

Description	Manage network security groups and their rules, and retrieve security lists.
Collection	`ansible.oci` (Certified, Red Hat)
Key content	**Network security groups** `oci_network_security_group`, `oci_network_security_group_info` **Security group rules** `oci_network_security_group_rule` **Security lists (management)** `oci_network_security_list` **Security list information** `oci_network_security_list_info`

Use Case: Balance load across backends

Description	Create load balancers with listeners, backend sets, backends, health checkers and routing policies.
Collection	`ansible.oci` (Certified, Red Hat)
Key content	**Load balancers** `oci_loadbalancer`, `oci_loadbalancer_info` **Listeners** `oci_loadbalancer_listener`, `oci_loadbalancer_listener_info` **Backend sets** `oci_loadbalancer_backend_set`, `oci_loadbalancer_backend_set_info` **Backends** `oci_loadbalancer_backend`, `oci_loadbalancer_backend_info` **Health checkers** `oci_loadbalancer_health_checker`, `oci_loadbalancer_health_checker_info` **Routing policies** `oci_loadbalancer_routing_policy`, `oci_loadbalancer_routing_policy_info`

Use Case: Manage object storage and databases

Description	Manage Object Storage buckets, objects and lifecycle policies, and provision Autonomous Databases.
Collection	`ansible.oci` (Certified, Red Hat)
Key content	**Buckets** `oci_object_storage_bucket`, `oci_object_storage_bucket_info` **Objects** `oci_object_storage_object`, `oci_object_storage_object_info` **Lifecycle policies** `oci_object_lifecycle_policy`, `oci_object_lifecycle_policy_info` **Autonomous Databases** `oci_autonomous_database`, `oci_autonomous_database_info`

Use Case: Run Kubernetes clusters on OKE

Description	Create OKE clusters and node pools, and look up available node pool options.
Collection	`ansible.oci` (Certified, Red Hat)
Key content	**OKE clusters** `oci_oke_cluster`, `oci_oke_cluster_info` **Node pools** `oci_oke_node_pool`, `oci_oke_node_pool_info` **Node pool options** `oci_oke_node_pool_options_info`

Use Case: Control access, compartments and tagging

Description	Manage compartments, identity domains, users, groups, dynamic groups, policies, API keys, tags and bastions for secure access.
Collection	`ansible.oci` (Certified, Red Hat)

Key content	**Compartments** `oci_identity_compartment`, `oci_identity_compartment_info` **Identity domains** `oci_identity_domain`, `oci_identity_domain_info` **Users** `oci_identity_user`, `oci_identity_user_info` **Groups** `oci_identity_group`, `oci_identity_group_info` **Group memberships** `oci_identity_user_group_membership`, `oci_identity_user_group_membership_info` **Dynamic groups** `oci_identity_dynamic_group`, `oci_identity_dynamic_group_info` **Policies** `oci_identity_policy`, `oci_identity_policy_info` **API keys** `oci_identity_api_key`, `oci_identity_api_key_info` **Availability domains** `oci_identity_availability_domain_info` **Tag namespaces** `oci_identity_tag_namespace`, `oci_identity_tag_namespace_info` **Tags** `oci_identity_tag`, `oci_identity_tag_info` **Bastions** `oci_bastion`, `oci_bastion_info`

OpenStack (workload migration) `os_migrate.os_migrate`

Certified Red Hat · version 1.0.5 · updated 2026-06-08 · 48 modules → 6 use cases

Use Case: Migrate projects, users and role assignments

Description	Export projects, users and their role assignments from a source OpenStack cloud and import them into a destination cloud.
Collection	`os_migrate.os_migrate` (Certified, Red Hat)
Key content	**Export project** `export_project` **Export user** `export_user` **Export user project role assignment** `export_user_project_role_assignment` **Import project** `import_project` **Import user** `import_user` **Import user project role assignment** `import_user_project_role_assignment` **Role assignments information** `os_role_assignments_info` **Export projects playbook** playbook `export_projects` **Export user project role assignments playbook** playbook `export_user_project_role_assignments` **Export users playbook** playbook `export_users` **Import projects playbook** playbook `import_projects` **Import user project role assignments playbook** playbook `import_user_project_role_assignments` **Import users playbook** playbook `import_users` **Export projects role** role `export_projects` **Export user project role assignments role** role `export_user_project_role_assignments` **Export users role** role `export_users` **Import projects role** role `import_projects` **Import user project role assignments role** role `import_user_project_role_assignments` **Import users role** role `import_users`
Notes	• Most resources have an export module, an import module, a playbook and a role of the same name.

Use Case: Migrate networks and security groups

Description	Export and import networks, subnets, routers and router interfaces, and security groups with their rules, and query routers and security groups.

Collection	`os_migrate.os_migrate` (Certified, Red Hat)
Key content	**Export network** `export_network` **Export router** `export_router` **Export router interfaces** `export_router_interfaces` **Export security group** `export_security_group` **Export security group rules** `export_security_group_rules` **Export subnet** `export_subnet` **Import network** `import_network` **Import router** `import_router` **Import router interface** `import_router_interface` **Import security group** `import_security_group` **Import security group rule** `import_security_group_rule` **Import subnet** `import_subnet` **Routers information** `os_routers_info` **Security groups information** `os_security_groups_info` **Export networks playbook** playbook `export_networks` **Export router interfaces playbook** playbook `export_router_interfaces` **Export routers playbook** playbook `export_routers` **Export security group rules playbook** playbook `export_security_group_rules` **Export security groups playbook** playbook `export_security_groups` **Export subnets playbook** playbook `export_subnets` **Import networks playbook** playbook `import_networks` **Import router interfaces playbook** playbook `import_router_interfaces` **Import routers playbook** playbook `import_routers` **Import security group rules playbook** playbook `import_security_group_rules` **Import security groups playbook** playbook `import_security_groups` **Import subnets playbook** playbook `import_subnets` **Export networks role** role `export_networks` **Export router interfaces role** role `export_router_interfaces` **Export routers role** role `export_routers` **Export security group rules role** role `export_security_group_rules` **Export security groups role** role `export_security_groups` **Export subnets role** role `export_subnets` **Import networks role** role `import_networks` **Import router interfaces role** role `import_router_interfaces` **Import routers role** role `import_routers` **Import security group rules role** role `import_security_group_rules` **Import security groups role** role `import_security_groups` **Import subnets role** role `import_subnets`
Notes	• Most resources have an export module, an import module, a playbook and a role of the same name.

Use Case: Migrate flavors, images and keypairs

Description	Export and import Nova flavors, Glance images and keypairs, including users' keypairs.
Collection	`os_migrate.os_migrate` (Certified, Red Hat)

Key content	**Export flavor** `export_flavor` **Export image blob** `export_image_blob` **Export image meta** `export_image_meta` **Export keypair** `export_keypair` **Import flavor** `import_flavor` **Import image** `import_image` **Import keypair** `import_keypair` **Keypairs information** `os_keypairs_info` **Export flavors playbook** playbook `export_flavors` **Export images playbook** playbook `export_images` **Export keypairs playbook** playbook `export_keypairs` **Export users keypairs playbook** playbook `export_users_keypairs` **Import flavors playbook** playbook `import_flavors` **Import images playbook** playbook `import_images` **Import keypairs playbook** playbook `import_keypairs` **Import users keypairs playbook** playbook `import_users_keypairs` **Export flavors role** role `export_flavors` **Export images role** role `export_images` **Export keypairs role** role `export_keypairs` **Export users keypairs role** role `export_users_keypairs` **Import flavors role** role `import_flavors` **Import images role** role `import_images` **Import keypairs role** role `import_keypairs` **Import users keypairs role** role `import_users_keypairs`
Notes	• Most resources have an export module, an import module, a playbook and a role of the same name.

Use Case: Migrate workloads and volumes

Description	Export instances and detached volumes, then copy their data to the destination with NBD exports and volume transfers, cleaning up temporary volumes afterwards.
Collection	`os_migrate.os_migrate` (Certified, Red Hat)
Key content	**Export detached volume** `export_detached_volume` **Export workload** `export_workload` **Import volumes export** `import_volumes_export` **Import volumes src cleanup** `import_volumes_src_cleanup` **Import volumes transfer** `import_volumes_transfer` **Import workload create instance** `import_workload_create_instance` **Import workload dst check** `import_workload_dst_check` **Import workload dst failure cleanup** `import_workload_dst_failure_cleanup` **Import workload export volume map** `import_workload_export_volume_map` **Import workload export volumes** `import_workload_export_volumes` **Import workload prelim** `import_workload_prelim` **Import workload src check** `import_workload_src_check` **Import workload src cleanup** `import_workload_src_cleanup` **Import workload transfer volumes** `import_workload_transfer_volumes` **Update workload nbdkit uris** `update_workload_nbdkit_uris` **Export detached volumes playbook** playbook `export_detached_volumes` **Export workloads playbook** playbook `export_workloads` **Import detached volumes playbook** playbook `import_detached_volumes` **Import from hypervisor playbook** playbook `import_from_hypervisor` **Import workloads playbook** playbook `import_workloads` **Export detached volumes role** role `export_detached_volumes` **Export workloads role** role `export_workloads` **Import detached volumes role** role `import_detached_volumes` **Import from hypervisor role** role `import_from_hypervisor` **Import workloads role** role `import_workloads`

Notes	• Most resources have an export module, an import module, a playbook and a role of the same name.

Use Case: Deploy conversion hosts and gather migration facts

Description	Deploy and delete the conversion hosts that move data, prepare the source and destination clouds, check the authenticated user and conversion host, and filter data in playbooks.
Collection	`os_migrate.os_migrate` (Certified, Red Hat)
Key content	**Authenticated user and project information** `auth_info` **Conversion host information** `os_conversion_host_info` **JMESPath query filter** filter `json_query` **String match filter** filter `stringfilter` **Delete conversion hosts playbook** playbook `delete_conversion_hosts` **Deploy conversion hosts playbook** playbook `deploy_conversion_hosts` **Conversion host role** role `conversion_host` **Conversion host content role** role `conversion_host_content` **Prelude common role** role `prelude_common` **Prelude dst role** role `prelude_dst` **Prelude src role** role `prelude_src`
Notes	• Most resources have an export module, an import module, a playbook and a role of the same name.

Use Case: Validate migration data

Description	Read exported resource files and validate the data directory and resource files before import.
Collection	`os_migrate.os_migrate` (Certified, Red Hat)
Key content	**Resource file reading** `read_resources` **Validate resource files** `validate_resource_files` **Validate data dir playbook** playbook `validate_data_dir` **Validate data dir role** role `validate_data_dir` **Validate resource files role** role `validate_resource_files`
Notes	• Most resources have an export module, an import module, a playbook and a role of the same name.